CHAPTER 23 · DERIVATIONS

GEOMETRICAL OPTICS

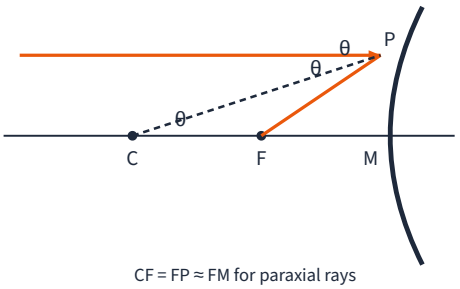
6 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 FOCAL LENGTH OF A SPHERICAL MIRROR

SYLLABUS 23.1(A)

RESULT $f = \frac{r}{2}$

1. A ray parallel to the axis strikes a concave mirror at P. The normal at P passes through the centre of curvature C.
2. Angle of incidence = angle of reflection = θ. The parallel ray makes angle θ with CP too (alternate angles), so triangle CFP has two equal angles.
3. CF = FP (isosceles triangle).
4. For paraxial rays P is close to the pole M, so FP ≈ FM. Then CM = CF + FM ≈ 2FM.
5. r = 2f, so $\mathbf{f = \frac{r}{2}}$.



CF = FP ≈ FM for paraxial rays

Use the sign convention consistently.

2 MIRROR EQUATION (REFLECTION AT A CURVED SURFACE)

SYLLABUS 23.1

RESULT $\frac{1}{u} + \frac{1}{v} = \frac{2}{r} = \frac{1}{f}$

1. A ray from a point object O on the axis strikes a concave mirror at P (height h above the axis) and reflects to the image I. The normal at P passes through the centre of curvature C.
2. Let the angles that OP, CP and IP make with the axis be α, β and γ. The angle of incidence is θ = β − α and the angle of reflection is θ = γ − β.
3. So β − α = γ − β, which gives α + γ = 2β.
4. Rays are very close to the principal axis (paraxial), so the angles are small: $\alpha \approx \frac{h}{u}, \beta \approx \frac{h}{r}, \gamma \approx \frac{h}{v}$.
5. $\frac{h}{u} + \frac{h}{v} = \frac{2h}{r}$, so $\mathbf{\frac{1}{u} + \frac{1}{v} = \frac{2}{r} = \frac{1}{f}}$.

Alternative (geometrical) method: use two pairs of similar triangles and relate their sides to the height h.

3 EQUATION EQUIVALENT TO SNELL'S LAW FOR PLANE SURFACES

SYLLABUS 23.2

RESULT $n_1 \sin\theta_1 = n_2 \sin\theta_2$

1. A ray in air at angle θ_a passes through two parallel-sided layers of refractive indices n_1 and n_2 and emerges into air again.
2. The surfaces are parallel, so the ray leaves into air at the same angle θ_a as it entered.
3. Air → medium 1: $n_1 = \dfrac{\sin\theta_a}{\sin\theta_1}$, so $\sin\theta_a = n_1 \sin\theta_1$.
4. Medium 2 → air: $n_2 = \dfrac{\sin\theta_a}{\sin\theta_2}$, so $\sin\theta_a = n_2 \sin\theta_2$.
5. Relate the angles: $\mathbf{n_1 \sin\theta_1 = n_2 \sin\theta_2}$ (n sin θ is constant across plane boundaries).

4 REFRACTION AT A CURVED SURFACE

SYLLABUS 23.2

RESULT $\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{r}$

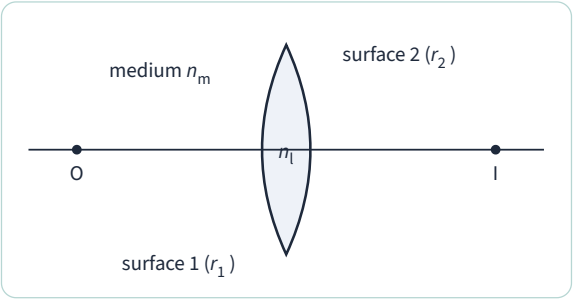
1. A ray from a point object O in medium n_1 strikes a convex spherical surface at P (height h) and refracts into medium n_2 to the image I. The normal at P passes through the centre of curvature C.
2. Let the angles that OP, CP and IP make with the axis be α, β and γ. Angle of incidence $\theta_1 = \alpha + \beta$; angle of refraction $\theta_2 = \beta - \gamma$.
3. Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$. For paraxial rays the angles are small: $n_1\theta_1 = n_2\theta_2$.
4. $n_1(\alpha + \beta) = n_2(\beta - \gamma)$, so $n_1\alpha + n_2\gamma = (n_2 - n_1)\beta$.
5. Small angles: $\alpha \approx \dfrac{h}{u}$, $\beta \approx \dfrac{h}{r}$, $\gamma \approx \dfrac{h}{v}$, so $\mathbf{\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{r}}$.

5 THIN LENS FORMULA AND LENSMAKER'S EQUATION

SYLLABUS 23.3(A)

RESULT $\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$; $\frac{1}{f_m}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})$

1. Lens of refractive index n_l in a medium n_m. Use $\frac{n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{r}$ at each surface.
2. Surface 1 (medium → lens): $\frac{n_m}{u}+\frac{n_l}{v_1}=\frac{n_l-n_m}{r_1}$ … (1)
3. The image I_1 is the object for surface 2. The lens is thin, so it is a virtual object at $u_2=-v_1$.
4. Surface 2 (lens → medium): $-\frac{n_l}{v_1}+\frac{n_m}{v}=\frac{n_m-n_l}{r_2}$ … (2)
5. Add (1) and (2): $\frac{n_m}{u}+\frac{n_m}{v}=(n_l-n_m)(\frac{1}{r_1}-\frac{1}{r_2})$.
6. Divide by n_m: $\frac{1}{u}+\frac{1}{v}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})$.
7. Object at infinity: $v=f_m$, so $\mathbf{\frac{1}{f_m}=(\frac{n_l}{n_m}-1)(\frac{1}{r_1}-\frac{1}{r_2})}$ (lensmaker's equation), and therefore $\mathbf{\frac{1}{u}+\frac{1}{v}=\frac{1}{f}}$.



In this form r is positive when the centre of curvature is on the side the light goes to, so for a biconvex lens r_2 is negative. Many books write $(\frac{1}{r_1}+\frac{1}{r_2})$ with both radii of a convex lens positive; both give the same f.

6 MINIMUM DISTANCE BETWEEN A REAL OBJECT AND ITS REAL IMAGE IS 4F

SYLLABUS 23.3

RESULT $d_{min}=4f$

1. Let the distance between the real object and the real image be d, so $u+v=d$ and $v=d-u$.
2. Substitute into the thin lens equation: $\frac{1}{u}+\frac{1}{d-u}=\frac{1}{f}$, so $d/(u(d-u))=\frac{1}{f}$.
3. Rearrange as a quadratic in u: $u^2-du+df=0$.
4. For a real image, u must be real: $b^2-4ac\geq 0$, so $d^2-4df\geq 0$ and $d\geq 4f$.
5. $\mathbf{d_{min}=4f}$, when $u=v=2f$.